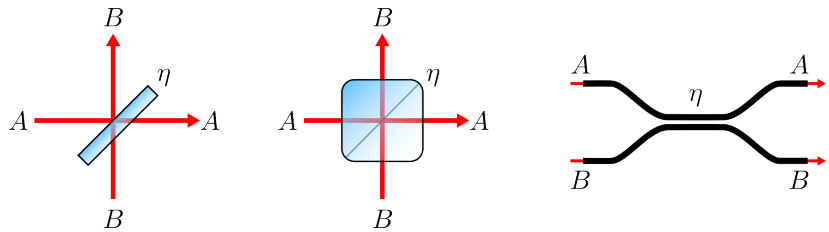


Figure 3.11: Different conventions for depicting beam splitters commonly seen in the scientific literature. *Left and middle:* free-space, bulk optics beam-splitters. *Right:* Fibre or wave-guide beam splitter.

they correspond to transformations of the annihilation operators of the form

$$\hat{a}_k \to U_{k1}\hat{a}_1 + U_{k1}\hat{a}_2 + \cdots + U_{k1}\hat{a}_N = \sum_{l=1}^{N} U_{kl}\hat{a}_l, \tag{3.133}$$

where the matrix $\boldsymbol{U}$ with elements $U_{kl}$ must be unitary. Eqivalently, collecting the annihilation operators in a vector $\hat{\boldsymbol{a}}$ then $\hat{\boldsymbol{a}} \to \boldsymbol{U}\hat{\boldsymbol{a}}$. The unitarity of $\boldsymbol{U}$ ensures that the canonical commutation relations are preserved. Furthermore, because (3.133) does not mix annihilation and creation operators, energy is conserved.

In practice, there are a lot of different physical components available that all implement transformations of the form (3.133) (at least up to small imperfections such as loss) and the same transformation may be implemented in various different ways. Which implementation is possible or preferred will depend on the practical setup. However, formally every passive linear-optical transformation can be constructed from just two basic operations: phase shifts and beam splitters.

### Phase shifts

Phase shifts have already been discussed in Sec. 3.1.4. They can be implemented by changing the optical path length of one mode relative to other modes in the system (only relative phases changes are observable). In free-space optics, this can be done by changing the spatial path length or by inserting an element with a different index of refraction, such as a piece of glass, into the path. In optical fibres, phase shifts can be achieved by applying tension, e.g. by slightly stretching the fibre.

### Beam splitters

Beam splitters act on a pair of optical modes. They allow some of the light from each mode to be transmitted or reflected into the other, creating interference (see Fig. 3.11). For two modes $A$, $B$ with annihilation operators $\hat{a}$ and $\hat{b}$, the